## Matrix-free first spectral coefficients by field resultants

### Prepared H,E0,E1 -> field Euclidean arithmetic

Monic square-free H; norm(E0+s E1) = A+s B, s^2=0

### Unit E0 -> resultant and trace coefficient

A=Res(H,E0); B=A*coeff_(b-1)((E1/E0)*H' mod H)

### One common root t -> smaller resultant

A=0; B=E1(t)*Res(H/(Z-t),E0); empty norm is one

### Two common roots -> zero first coefficient

No nilpotent inverse; chosen norm feeds block output, with dense and stream references

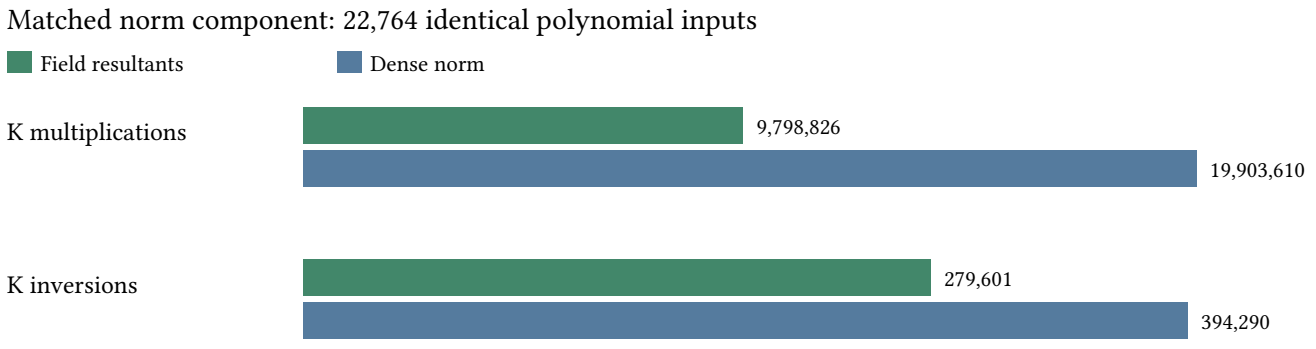
Construction, 2026-11-09, T-1947. The signed baby-algebra norm now has a field-only first-coefficient routine with explicit unit, simple-common-root and multiple-root branches. It avoids constructing a multiplication matrix, preserves the marked coefficient when the constant norm is zero, and explicitly supplies the selected norm to the complete-block response. All 381 prior decoder cases remain byte-identical. Written proofs and native checks remain unverified producer evidence. [pcc:unverified]

Three algebraic cases. For square-free monic H and Norm(E0+s*E1)=A+s*B, a unit E0 gives A=Res(H,E0) and B=A*coeff_(b-1)((E1/E0)*H' modH). If the common divisor is Z-t, then A=0 and B=E1(t)*Res(H/(Z-t),E0). Two or more common roots give A=B=0. Zero perturbation uses the ordinary resultant directly. The proof derives these identities by roots and Lagrange interpolation, including the degree-one empty-norm edge. Every inverse is in the field; no zero or nilpotent norm is inverted. [pcc:unverified]

Implemented arithmetic. The code uses classical Euclidean polynomial division, resultants and extended gcd. It is not a subresultant PRS, half-gcd or fast polynomial package. At this implementation level the isolated norm costs O(b^2) field work and linear polynomial workspace, versus cubic dense elimination and quadratic matrix storage. The qualifier retains both backends, so total process work and memory also include the dense reference. Input-polynomial preparation, interpolation, points, calibration and decoding are separate costs. [pcc:unverified]

Complete exact panel. The 35 synthetic norm controls agree under field resultants, dense unit-pivot determinants and explicit root products, covering units, one common root, multiple common roots and zero E0. The unchanged decoder panel has 381 cases and 2694 coefficient queries. Its 11382 paired blocks supply 22764 numerator/denominator norm inputs:9612 unit cases,1770 simple-common-root cases, zero multiple-root cases and 11382 zero-perturbation denominator cases. The full scientific output is byte-identical to Round 63. [pcc:unverified]

Matched norm costs. Across those 22764 already-formed inputs in the three declared fields, the field-only routine records 9,798,826 quadratic-field multiplications and 279,601 inversions; the dense norm records 19,903,610 multiplications and 394,290 inversions. Multiplication counters include inversion powers. These are isolated deterministic counts on matching inputs, not timings or a complete response speedup. Raw per-query, preparation, stream, decoder and whole-qualifier receipts remain available. The selected norm value is fed into the complete-block response only after equality with the paired-stream reference. [pcc:unverified]




Source: response.log. Separate scales; multiplications include inverse powers. No preparation or timing claim.

Primary-curve checks. Full-width P192 arithmetic interpolates four signed pair jets for each of three blocks in the 24 existing 27-factor cases. All 72 field-resultant norms agree with the direct four-pair contraction:54 unit and 18 simple-root cases. Their 24 combined first coefficients match the old Round 63 file. Isolated P192 norms use 11,106 explicit K-multiplication and 1008 K-inverse calls; inverse internals and polynomial-input preparation have separate accounting. The inputs here are deliberately built from explicit factors. This qualifies the norm backend and does not supply a full P192 response or arbitrary-input recovery. [pcc:unverified]

Zero-count clarification. Round 63 reported 2190 complete zero blocks, which include their center factors. The new branch instrumentation distinguishes 1770 singular baby norms from 420 blocks that vanish only at the separate center factor. The old data is preserved; its 2190 count must not be read as 2190 singular b-by-b baby multiplication matrices. [pcc:unverified]

Replays and actual failures. The first build rejected an included helper under src/bin as an invalid extra executable. The corrected source moved it to src and deleted the old Git-tree path before qualification. A later adapter amendment made the selected norm feed the response explicitly; final-source replays reproduce four scientific outputs and both operation receipts byte-for-byte. Changed control and P192 records reject 101. The legacy library timeout 124 from Round 63 is retained and not retried without a changed condition; it is not a publication-head pass. The generic freeze receipt's stale PR number is annotated; the actual branch extended is PR #3767. [pcc:unverified]

Publication and next construction. The existing PR picked up an upstream append conflict. Its published Round 63 records and immutable PCC claim are preserved while merging canonical upstream state, and this rung extends the same PR. Next, replace each denominator norm by H_G(x(A))^4*H_Q(x(B))^4, using the already-known signed baby x-coordinates, then investigate sharing numerator work across giant blocks. This targets the remaining field work rather than adding another point-action oracle. No cloud experiment, generator or old freeze lane was started or modified. [pcc:unverified]

Sources checked. Round 63 supplies the actual signed algebra and frozen panel. Stacks 0BIE, https://stacks.math.columbia.edu/tag/0BIE , defines trace/norm by multiplication matrices; the etale-algebra formulas here are proved directly. BFLS, https://arxiv.org/pdf/2003.10118 , provides field-resultant and jet background. KB tools/wrapper remain unavailable. No novelty or prior-art absence claim is made for the standard trace, gcd or resultant ingredients. [pcc:unverified]